# INTRODUCTION TO BIG DATA

# Introduction to Big Data



- Big Data (What? Why? Where?)
- Big Data (Characteristics, Challenges & Applications)
- Enabling Technologies for Big Data
- Big Data Stack
- Big Data distribution packages

# WHAT IS BIG DATA?

# Big Data

Big Data is the term for a collection of datasets so **large** and **complex** that it becomes difficult to process using in-hand database management tools or traditional data processing applications.

--Wikipedia

The Basic idea behind the phrase "Big Data" is that everything we do is increasingly leaving a digital trace (or data), which we can use and analyze to become smarter. The driving forces in this brave new world of access to ever-increasing volumes of data and our ever-increasing technological capability to mine that data for commercial insights.

--Bernard Marr

"Big Data is high-volume, high-velocity and/or high variety information assets that demand cost-effective, Innovative forms of information processing that enable enhanced insights, decision making and process automation."

--Gartner

# Big Data

"Big Data refers to the dynamic, large and disparate volumes of data being created by people, tools and machines; it requires new, innovative and scalable technologies to collect, host and analytically process the vast amount of data gathered in order to derive real-time business insights that relate to consumers, risk, profit, performance, productivity management and enhanced shareholder value"

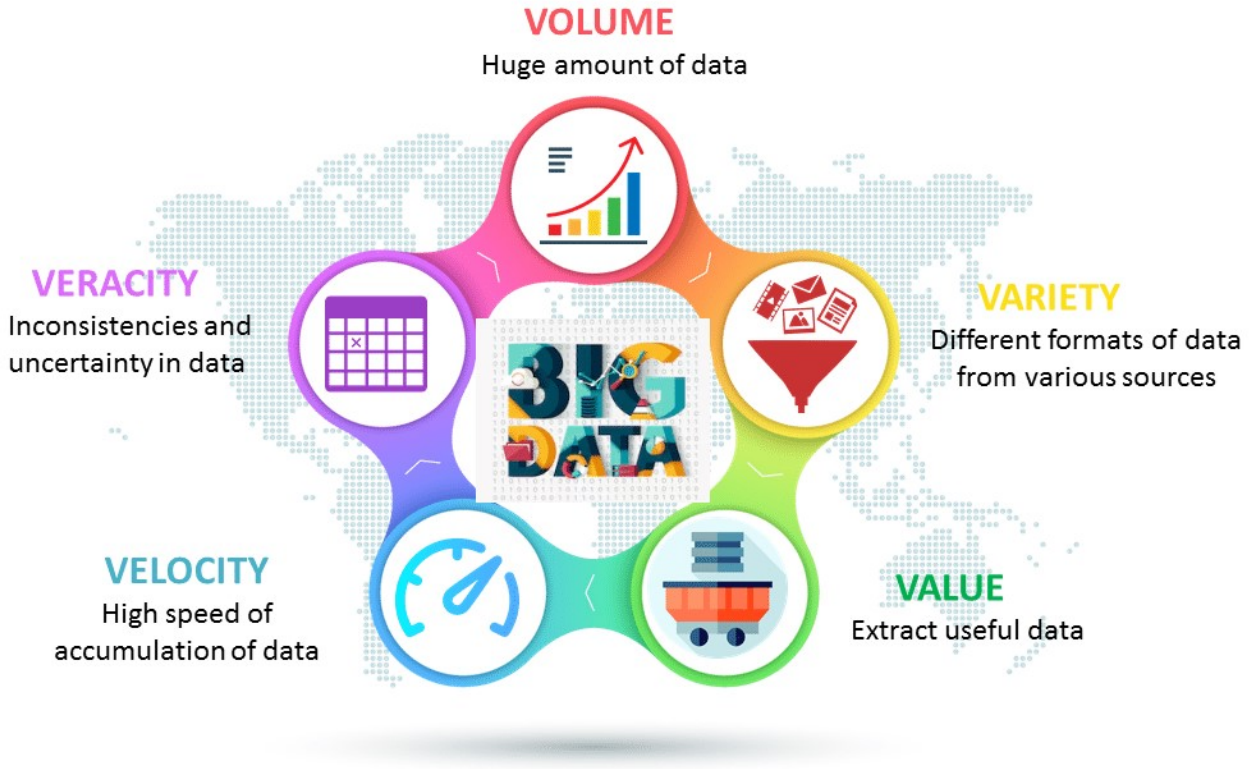
-- Ernst and Young

"Big Data is a collection of data from traditional and digital sources inside and outside your company that represents a source for ongoing discovery and analysis"

--- Lisa Arthur (Forbes Contributor)

# CHARACTERISTICS OF BIG DATA

# Characteristics of Big Data



# 1. Volume

**Definition**: Volume refers to the vast amounts of data generated every second from various sources such as social media, sensors, transactions, videos, and more.

- **Scale**: The sheer scale of data that organizations must process is enormous. This can range from terabytes to petabytes and beyond.
- **Storage**: Handling such large volumes of data requires advanced storage solutions like distributed storage systems (e.g., Hadoop HDFS).
- **Examples**: Social media posts, transaction records, sensor data from IoT devices, and multimedia content.

# 2. Velocity

**Definition**: Velocity refers to the speed at which data is generated, collected, and processed.

- **Real-Time Processing**: The ability to process data in real-time or near real-time is crucial for timely decision-making.
- **Streaming Data**: Technologies like Apache Kafka, Apache Storm, and Apache Flink are used to handle streaming data.
- **Examples**: Real-time stock trading data, social media updates, and online purchase transactions.

# 3. Variety

**Definition**: Variety refers to the different types of data that are generated from various sources.

- **Data Types**: Structured data (e.g., databases), semi-structured data (e.g., XML, JSON), and unstructured data (e.g., text, images, videos).
- **Integration**: Combining and analyzing diverse data types from different sources can provide more comprehensive insights.
- **Examples**: Text documents, emails, videos, images, audio files, and sensor data.
- .

# 4. Veracity

**Definition**: Veracity refers to the trustworthiness and quality of the data.

- **Data Quality**: Ensuring data accuracy, completeness, and consistency is essential for reliable analysis.
- **Uncertainty and Noise**: Big data often contains uncertainties, biases, and noise that must be managed and minimized.
- **Techniques**: Data cleaning, validation, and enrichment processes help improve data quality.
- **Examples**: Inaccurate sensor readings, social media rumors, and incomplete transaction records.

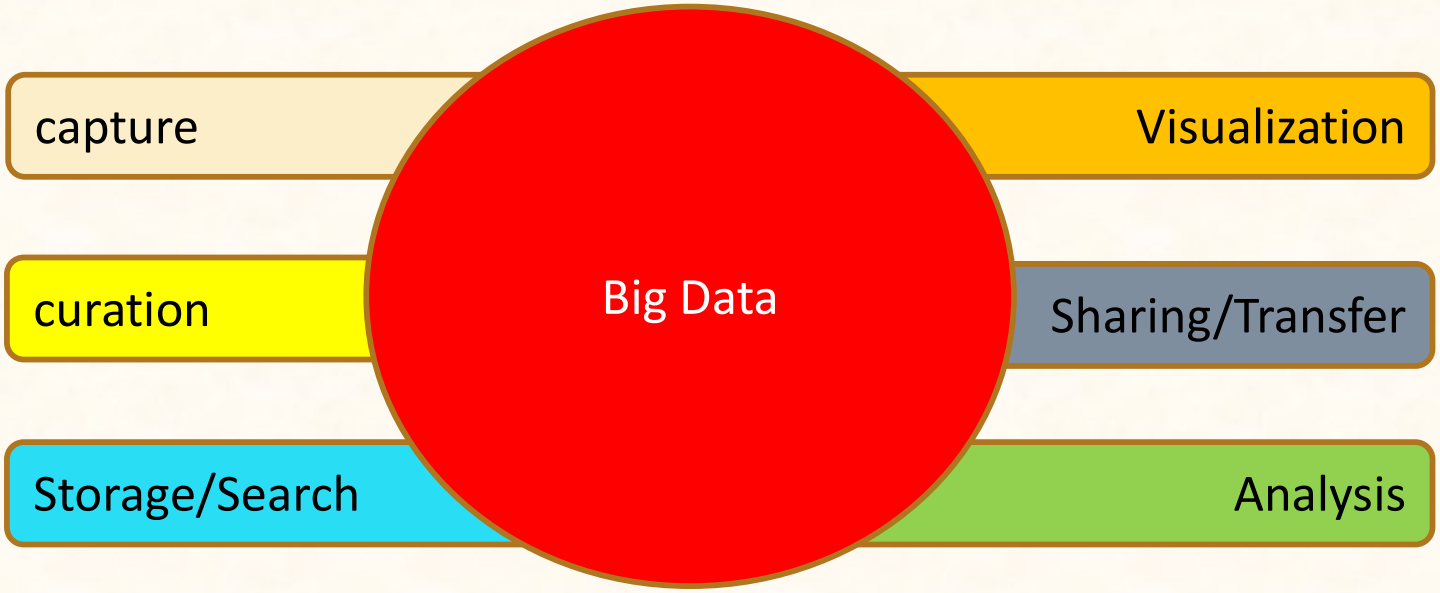
# 5. Value

**Definition**: Value refers to the potential of big data to generate meaningful insights and drive business value.

- **Actionable Insights**: Extracting valuable insights from big data can lead to better decision-making and competitive advantages.
- **Monetization**: Organizations can monetize data by creating new revenue streams or improving existing products and services.
- **Examples**: Predictive analytics for customer behavior, personalized marketing campaigns, and optimizing supply chain operations.

# CHALLENGES OF BIG DATA

# Challenges of Big Data



- **Capture**:
- The process of collecting and ingesting data from various sources.
- **Curation**:
- Organizing, integrating, and maintaining the collected data to ensure its quality and accessibility.
- **Storage/Search**:
- Efficiently storing large datasets and providing robust search capabilities to retrieve relevant data quickly.

- **Visualization**:
- Representing data in graphical formats to make the information more understandable and actionable.
- **Sharing/Transfer**:
- Ensuring data can be securely and efficiently shared or transferred between different systems or stakeholders.
- **Analysis**:
- Using various techniques and tools to analyze the data and extract valuable insights

**1. Data Storage and Management**

**Challenge**: Storing and managing the massive volumes of data generated daily is a critical challenge.

- **Scalability**: Ensuring storage solutions can scale to accommodate growing data volumes.
- **Infrastructure Costs**: The cost of setting up and maintaining large-scale storage systems can be high.
- **Data Retention**: Deciding which data to keep and for how long, given the storage constraints.

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**2. Data Processing and Analysis**

**Challenge**: Processing and analyzing large datasets efficiently and effectively.

- **Real-Time Processing**: Handling data in real-time or near real-time requires advanced processing capabilities.
- **Complexity**: Analyzing diverse data types (structured, semi-structured, unstructured) requires sophisticated tools and techniques.
- **Performance**: Ensuring data processing systems are fast and reliable enough to handle the workload.

**3. Data Integration**

**Challenge**: Integrating data from various sources and formats.

- **Heterogeneous Data**: Combining data from different sources (e.g., databases, social media, IoT devices) can be complex.
- **Data Silos**: Overcoming organizational data silos to create a unified view of the data.
- **Standardization**: Developing standards for data formats and protocols to facilitate integration.

**4. Data Quality and Veracity**

**Challenge**: Ensuring the accuracy, consistency, and reliability of data.

- **Data Cleaning**: Identifying and correcting errors, inconsistencies, and inaccuracies in the data.
- **Data Governance**: Implementing policies and procedures to maintain data quality over time.
- **Uncertainty and Noise**: Managing and minimizing uncertainties, biases, and noise in the data.

**5. Security and Privacy**

**Challenge**: Protecting sensitive data and ensuring privacy.

- **Data Breaches**: Preventing unauthorized access and data breaches.
- **Compliance**: Adhering to regulations and standards (e.g., GDPR, CCPA) regarding data privacy and protection.
- **Encryption**: Implementing strong encryption methods to protect data at rest and in transit.

**6. Talent and Skill Shortage**

**Challenge**: Finding and retaining skilled professionals who can manage and analyze big data.

- **Expertise**: The demand for data scientists, data engineers, and other specialized roles often outstrips supply.
- **Training**: Continuous training and development are needed to keep up with evolving technologies and methodologies.
- **Interdisciplinary Skills**: Combining knowledge of data science, domain expertise, and business acumen.

**7. Cost Management**

**Challenge**: Managing the costs associated with big data infrastructure and operations.

- **Budgeting**: Balancing the budget for big data projects with other organizational priorities.
- **Cost-Benefit Analysis**: Demonstrating the value and return on investment (ROI) of big data initiatives.
- **Operational Costs**: Controlling the operational costs of data storage, processing, and maintenance.

**8. Data Governance and Compliance**

**Challenge**: Establishing effective data governance frameworks and ensuring compliance with legal requirements.

- **Policies and Procedures**: Developing and enforcing policies for data access, usage, and management.
- **Regulatory Compliance**: Staying compliant with evolving regulations and standards across different jurisdictions.
- **Auditability**: Ensuring data processes are transparent and can be audited for compliance purposes.

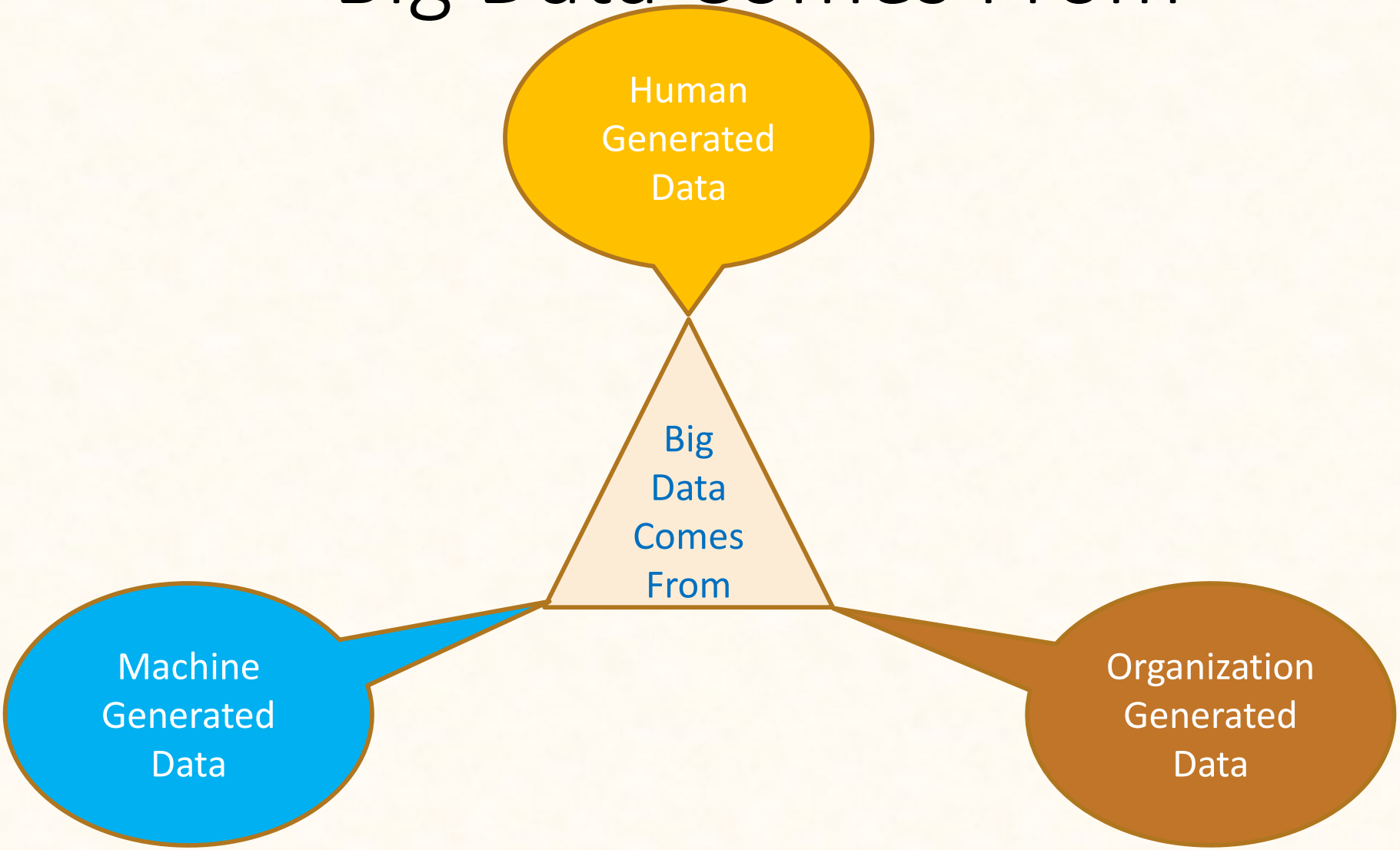
**9. Data Visualization and Interpretation**

**Challenge**: Presenting data in a way that is easily understandable and actionable for stakeholders.

- **Visualization Tools**: Selecting and using the right tools and techniques for effective data visualization.
- **Storytelling**: Translating complex data insights into compelling narratives that drive decision-making.
- **User Engagement**: Ensuring visualizations are interactive and engaging for users with varying levels of data literacy

# WHERE DID BIG DATA COMES FROM?

# Big Data Comes From

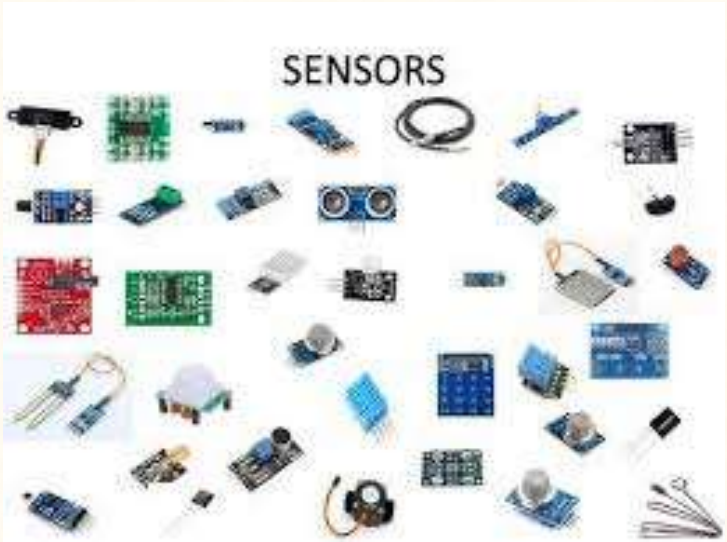


# Machine Generated Data







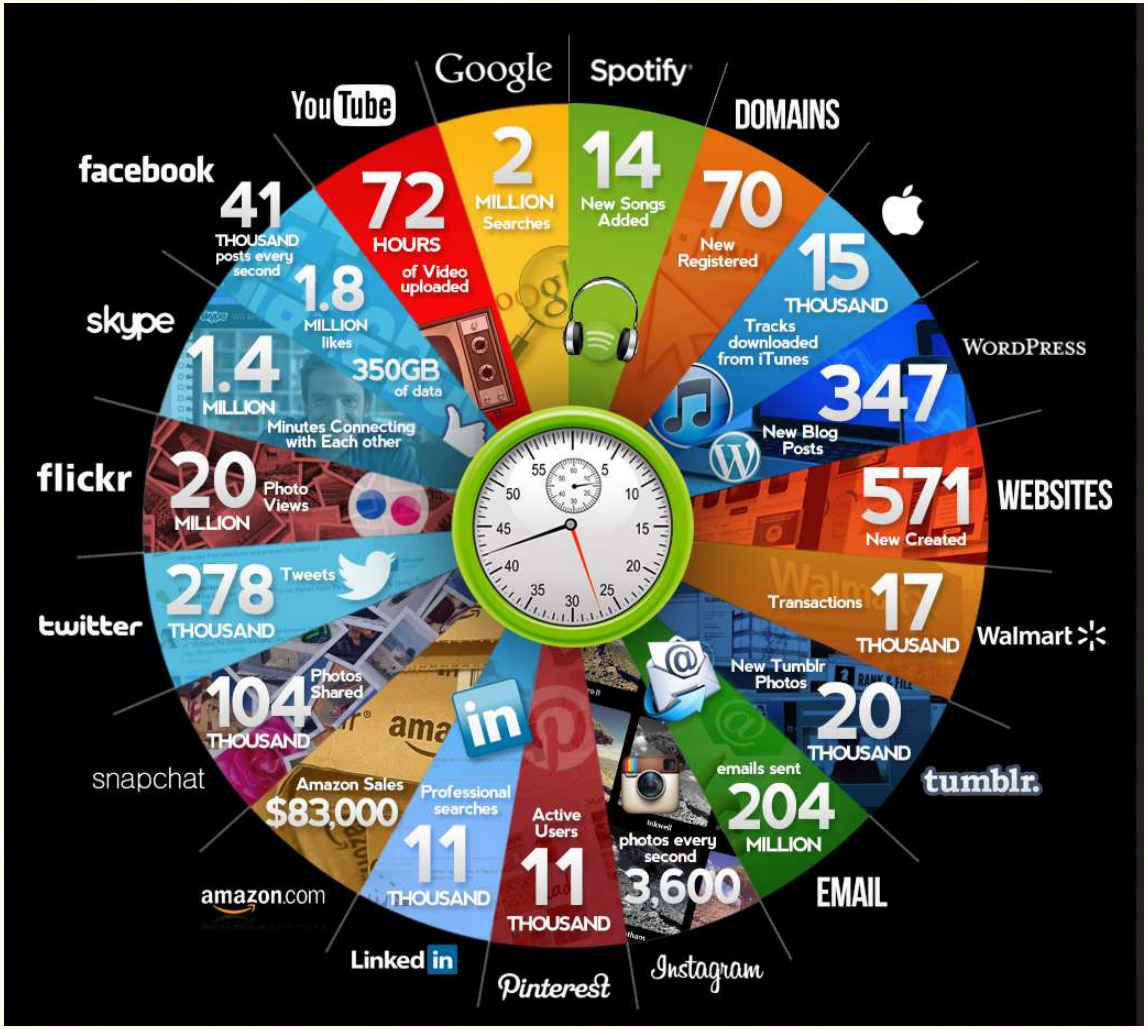






**Sensing → Smart**

# Human Generated Data




https://blog.qmee.com/wp-content/uploads/2013/07/Qmee-Online-In-60-Seconds2.png

# Organization Generated Data

- Commercial Transactions
- Banking/Stock Records
- Credit Cards
- Government Open Data
- E-Commerce
- Medical Records

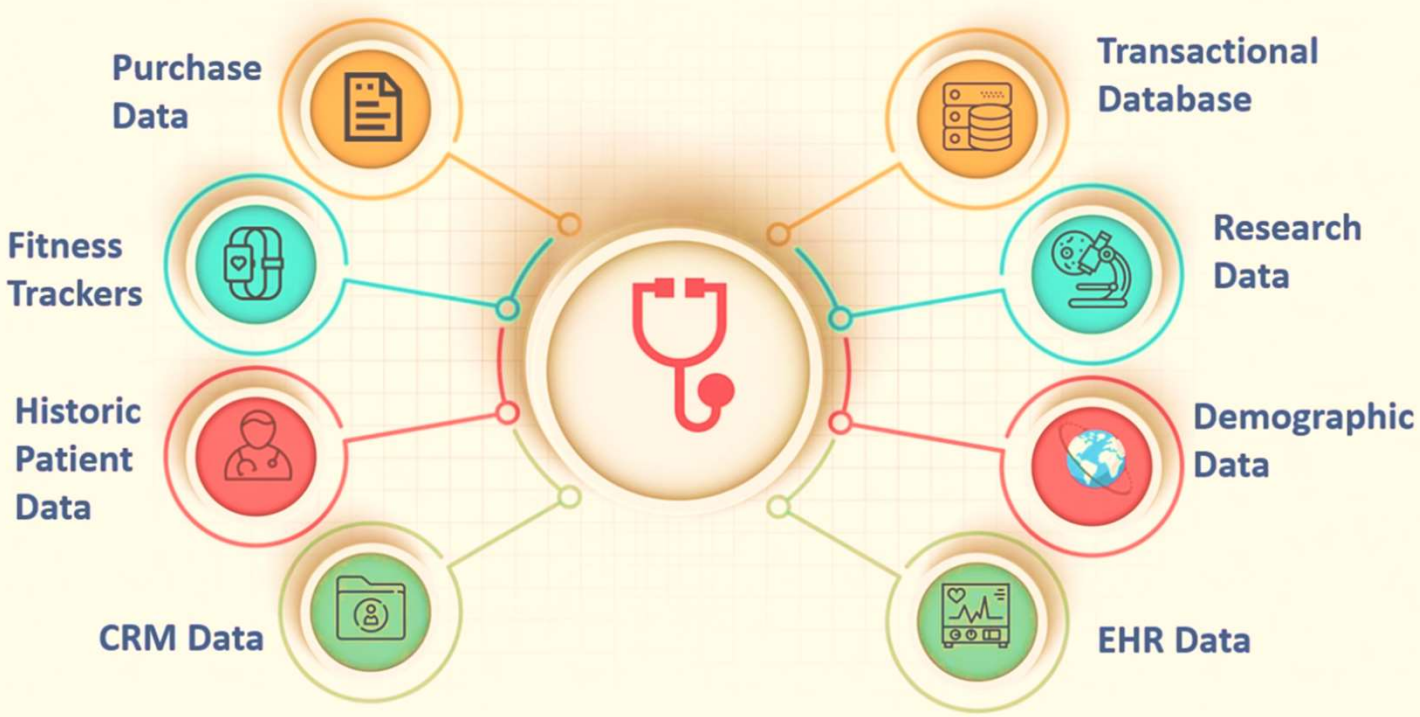
# APPLICATIONS OF BIG DATA

# Application of Big Data

- Health Care
- IOT
- Media and Entertainment
- Education
- E-Commerce
- Telecommunications
- Government
- Transportation & Hotels and Food Delivery Services

# Applications of Big Data

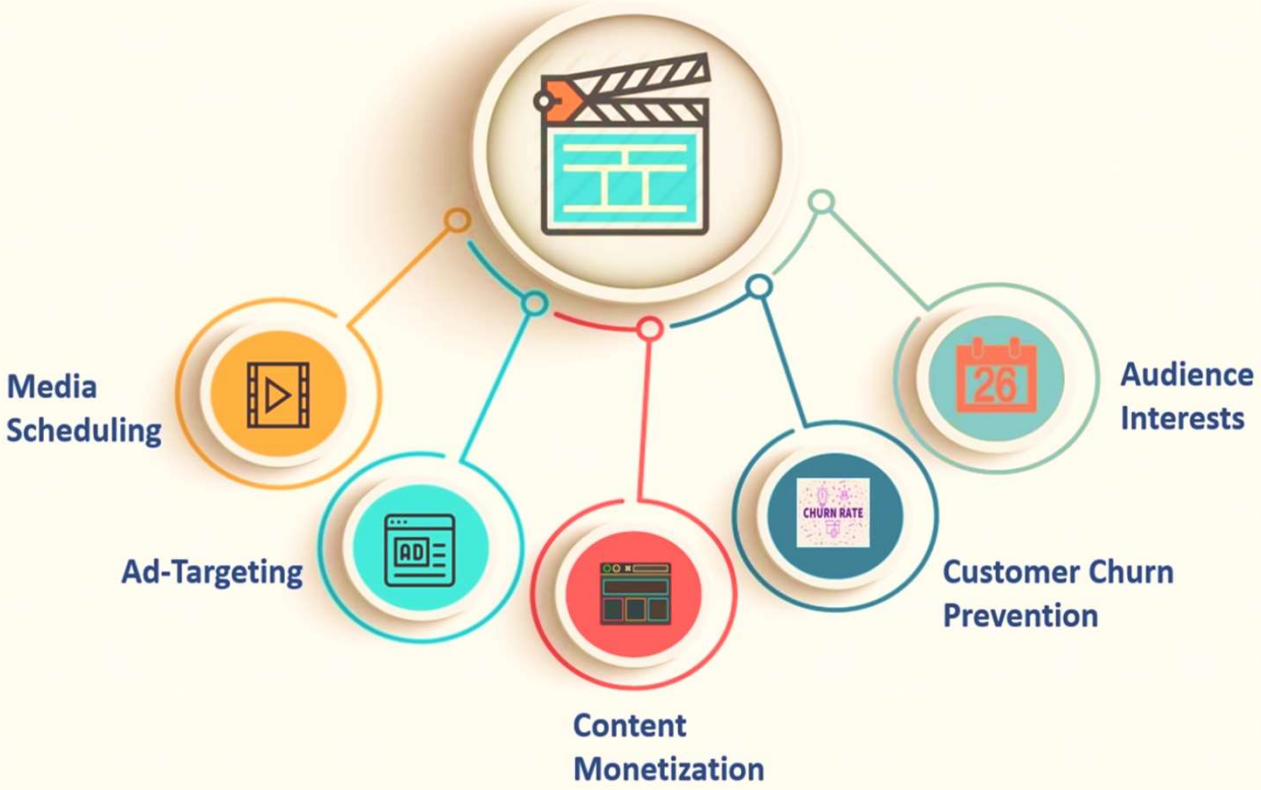
- Health Care
- Media and Entertainment
- E-Commerce
- Government
- IOT
- Education
- Tele-Communications
- Transportation, Hotels & Food Delivery Services




- Health Information Exchange
- Gene Sequencing / Medical research
- Personalized Treatment
- Cost Reduction

# Applications of Big Data

- Health Care
- Media and Entertainment
- E-Commerce
- Government
- IOT
- Education
- Tele-Communications
- Transportation, Hotels & Food Delivery Services




- Sentiment Analysis
- Customers Behaviour
- Recommendation Engines

# Applications of Big Data

- Health Care
- Media and Entertainment
- **E-Commerce**
- Government
- IOT
- Education
- Tele-Communications
- Transportation, Hotels & Food Delivery Services




- Sentiment Analysis
- Ad-Targeting
- Search Quality
- Point of Sales Transactional Analysis
- Customer Churn Analysis

# Applications of Big Data

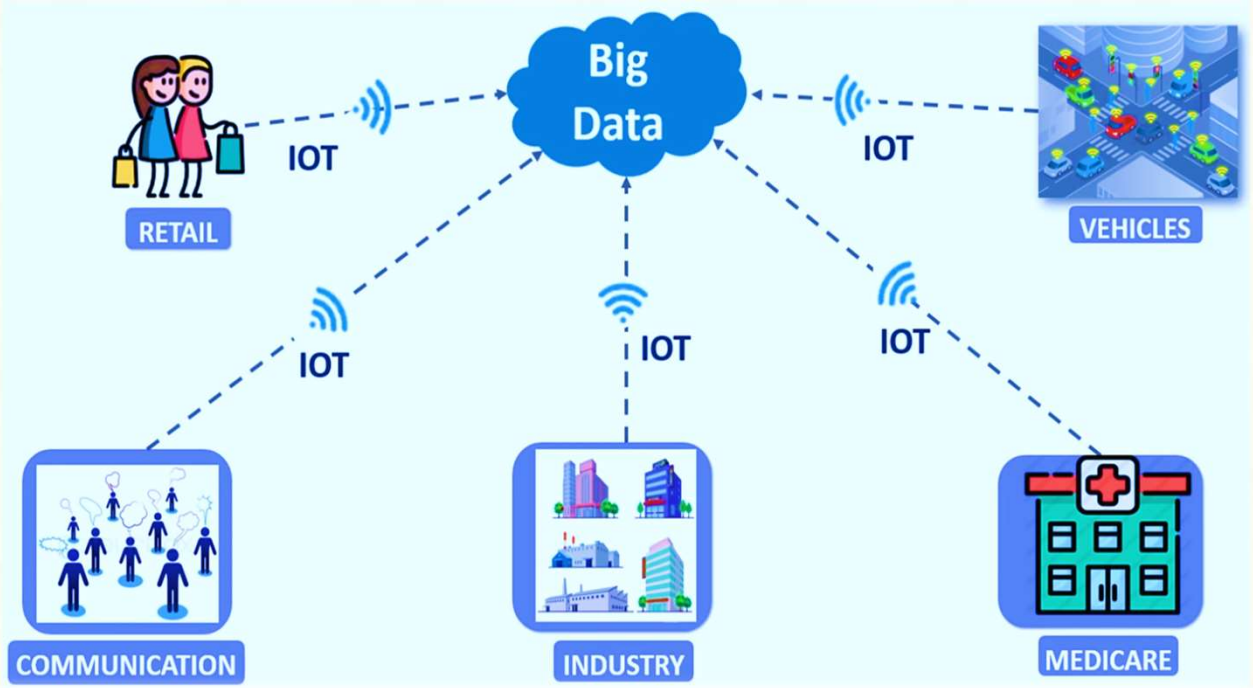
- Health Care
- Media and Entertainment
- E-Commerce
- **Government**
- IOT
- Education
- Tele-Communications
- Transportation, Hotels & Food Delivery Services




- Welfare Schemes (Improving social and economic policies)
- Justice to community
- Track unemployment
- Poverty

# Applications of Big Data

- Health Care
- Media and Entertainment
- E-Commerce
- Government
- **IOT**
- Education
- Tele-Communications
- Transportation, Hotels & Food Delivery Services




- Smart Home
- Smart Argriculture
- Smart Mobility
- Smart Energy (Electricity)

# Applications of Big Data

- Health Care
- Media and Entertainment
- E-Commerce
- Government
- IOT
- **Education**
- Tele-Communications
- Transportation, Hotels & Food Delivery Services

- Monitoring Individual Student Performance
- Preventing Attrition from a course or a program
- Identifying outline for early intervention
- Testing and evaluation of curriculum



# Applications of Big Data

- Health Care
- Media and Entertainment
- E-Commerce
- Government
- IOT
- Education
- **Tele-Communications**
- Transportation, Hotels & Food Delivery Services



- Customer Churn Prevention
- Network Performance Optimization
- Calling Data Record (CDR) Analysis
- Analysing Network to predict failures

# Applications of Big Data

- Health Care
- Media and Entertainment
- E-Commerce
- Government
- IOT
- Education
- Tele-Communications
- **Transportation, Hotels & Food Delivery Services**

## Transportation

- Surge Pricing
- Ride Pooling
- Revenue Management
- Traffic Control
- Route Planning
- Logistics

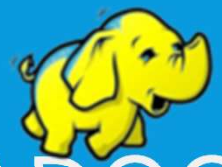
## Hostels & Food Delivery Services

- Customer Demands
- Details of Customers
- Availability and Seasonal Data Changes

# ENABLING TECHNOLOGIES FOR BIG DATA

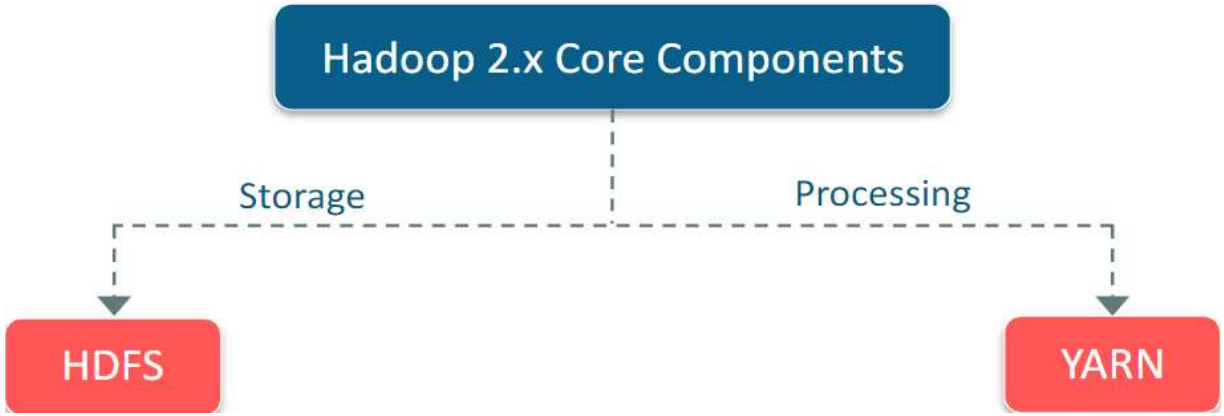
# What is Big Data Technology?

Big data technology can be defined as a software-utility that is designed to **analyze, process** and **extract** the information from an **extremely complex and large data sets** which the **traditional data processing software** could never deal with.



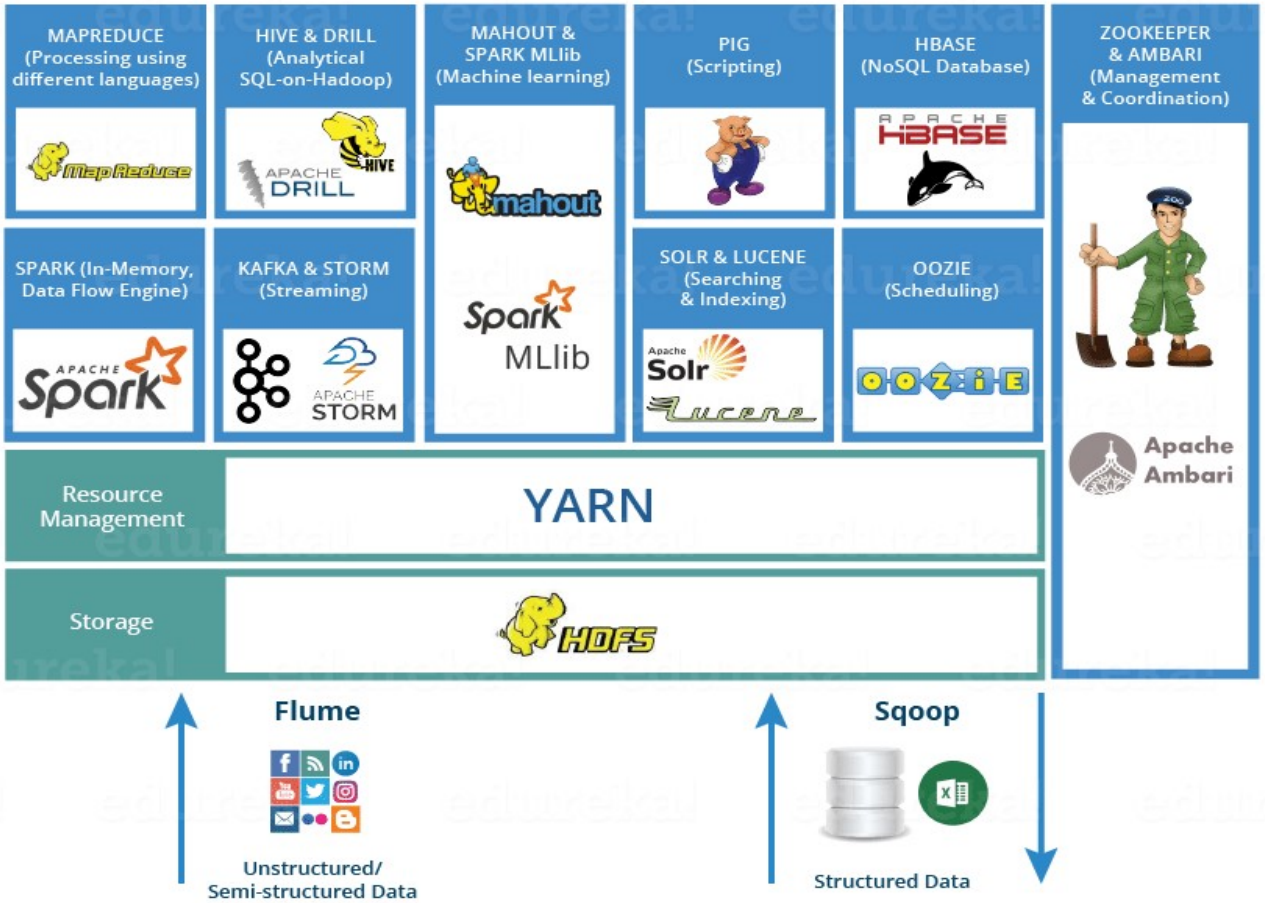
HADOOP

- Apache Hadoop is a framework that allows **storage** and **processing** of huge data sets across the Hadoop cluster in a **distributed** and **parallel** fashion.
- Hadoop cluster is **highly scalable** and uses **commodity hardware** which are relatively cheaper.




- **Developed by**: Apache Software Foundation in the year 2011 10th of Dec.
- **Written in**: JAVA









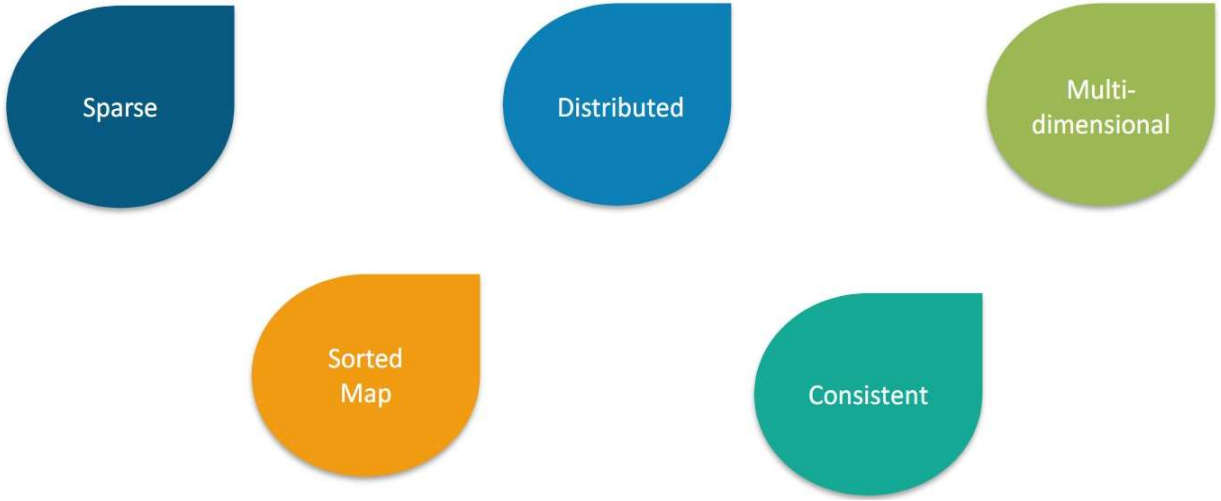
# HADOOP ECO-SYSTEM

# HADOOP ECO-SYSTEM

- HDFS -> *Hadoop Distributed File System*
- YARN -> *Yet Another Resource Negotiator*
- MapReduce -> *Data processing using programming*
- Spark -> In-memory Data Processing
- PIG, HIVE-> *Data Processing Services using Query (SQL-like)*
- HBase -> *NoSQL Database*
- Mahout, **Spark MLlib** -> *Machine Learning*
- Apache Drill -> *SQL on Hadoop*
- Zookeeper -> *Managing Cluster*
- Oozie -> *Job Scheduling*
- Flume, Sqoop -> *Data Ingesting Services*
- Solr & Lucene -> *Searching & Indexing*
- Ambari -> *Provision, Monitor and Maintain cluster*




- Hbase is non-relational (NoSQL) database which stores data in key-value pair.
- Specifically, it is:






# PIG

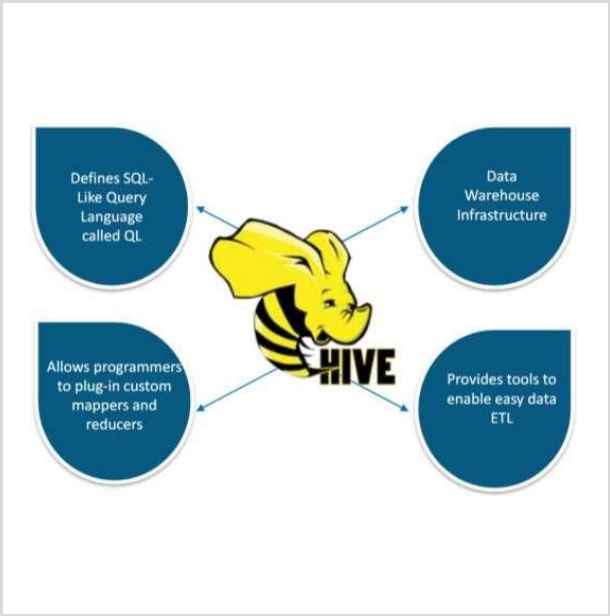
- Pig is an Open-Source high-level dataflow system.
- Pig uses Pig Latin language for Querying and Data Manipulation.
- Internally Pig commands are translated into Map-Reduce.

Why is it Important?

- Companies such as Yahoo, Google and Microsoft are collecting enormous data sets in the form of click streams, search logs and web crawls.
- Some form of ad-hoc processing and analysis of all this information is required.



- Hive One of the biggest ingredients in the Information Platform built by Jeff Hammerbacher team at Facebook was Hive, a framework for data warehousing on top of Hadoop.



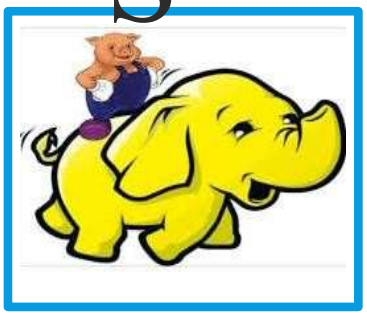






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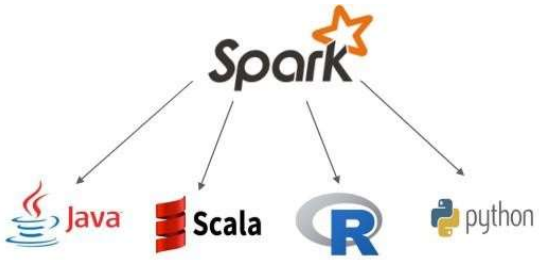
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| Features | Hive | Pig |
|---|---|---|
| Language | SQL-like | PigLatin |
| Schemas/Types | Yes (explicit) | Yes (implicit) |
| Partitions | Yes | No |
| Server | Optional (Thrift) | No |
| User Defined Functions (UDF) | Yes (Java) | Yes (Java) |
| Custom Serializer/Deserializer | Yes | Yes |
| DFS Direct Access | Yes (implicit) | Yes (explicit) |
| Join/Order/Sort | Yes | Yes |
| Shell | Yes | Yes |
| Streaming | Yes | Yes |
| Web Interface | Yes | No |
| JDBC/ODBC | Yes (limited) | No |



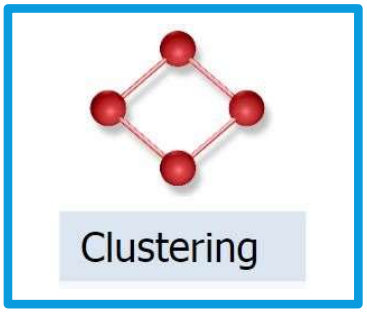

- As the name suggests, Apache Drill is used to drill into any kind of data.
- It's an **open source application** which works with **distributed environment** to **analyze** large data sets.
  - It is a replica of **Google Dremel.**
  - It supports different kinds NoSQL databases and file systems, which is a powerful feature of Drill. For example: **Azure Blob Storage, Google Cloud Storage, HBase, MongoDB, MapR-DB HDFS, MapR-FS, Amazon S3, Swift, NAS and local files.**
- So, basically the main aim behind Apache Drill is to **provide scalability** so that we can process petabytes and exabytes of data efficiently.
- It has a powerful scalability factor in supporting millions of users and serve their query requests over large scale data.
- The main power of Apache Drill lies in combining a **variety of data stores** just by using a single query.
- Apache Drill basically follows the **ANSI SQL**.

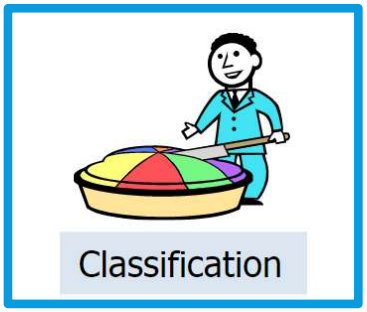



- Apache Spark is a framework for real time data analytics in a distributed computing environment.
- An in-memory cluster computing platform
- An in-memory cluster computing platform
- Runs lightning fast by performing computations in memory.
- Provides rich set of tools like
  - Spark SQL
  - Spark Streaming
  - GraphX
  - Mlib
- Runs on Windows and UNIX systems

Developed by Matei Zaharia at UC Berkeley Lab in 2009 and Open-Sourced in 2010 Written in functional Programming language Scala.




- ✓ Apache Mahout is an Apache project to produce open source implementations of distributed or otherwise Scalable Machine Learning Algorithms focused primarily in the areas of Collaborative Filtering, Clustering and Classification, often leveraging, but not limited to, the Hadoop platform.

- ✓ The Apache Mahout project aims to make building intelligent applications easier and faster. Mahout co-founder Grant Ingersoll introduces the basic concepts of machine learning and then demonstrates how to use Mahout to cluster documents, make recommendations, and organize content.

Three specific machine-learning tasks that Mahout currently implements are:

- ✓ Collaborative Filtering
- ✓ Clustering
- ✓ Classification

# SQOOP

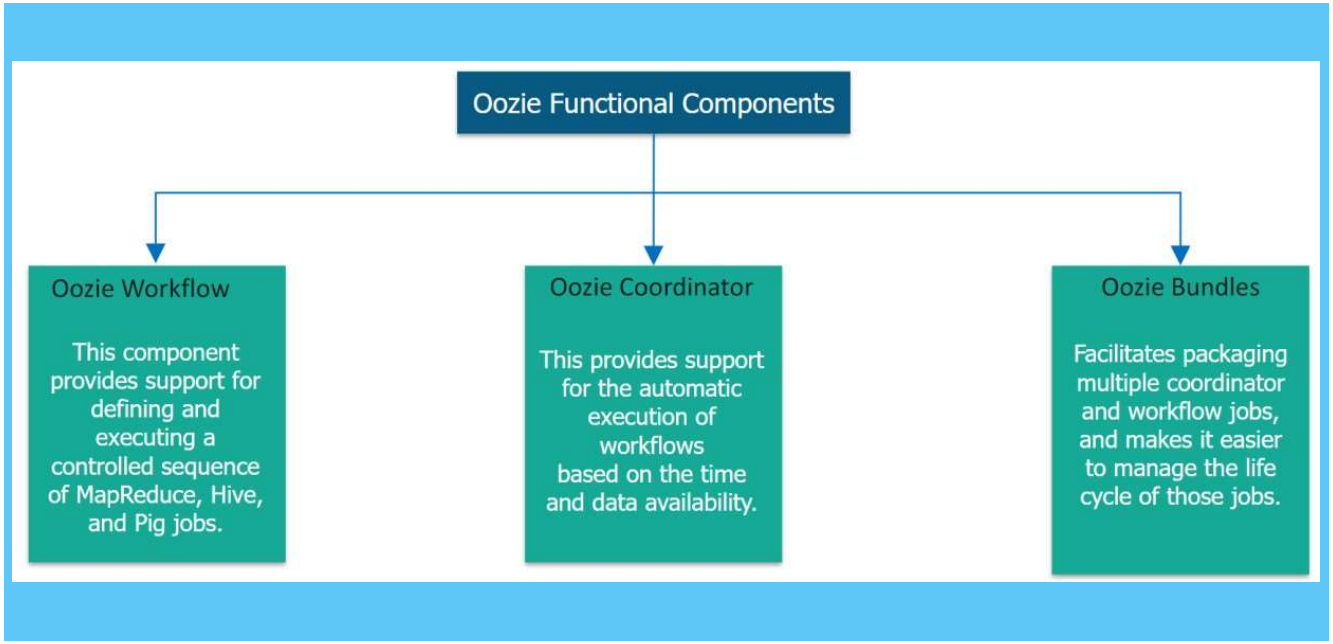
- Apache Sqoop is a tool which is designed to transfer data between HDFS (Hadoop storage) and relational database servers like MySQL, Oracle RDB, SQLite, Teradata, Netezza, Postgres etc.
- Apache Sqoop imports data from relational databases to HDFS, and exports data from HDFS to relational databases.
- **This is how Sqoop got its name – "SQL to Hadoop & Hadoop to SQL".**

# FLUME




- Apache Flume is a tool for data ingestion in HDFS.
- Flume captures streaming data from various web servers and transfers to HDFS.
- Large amount of streaming data such as log files, events from various sources like network traffic, social media, email messages etc. to HDFS.

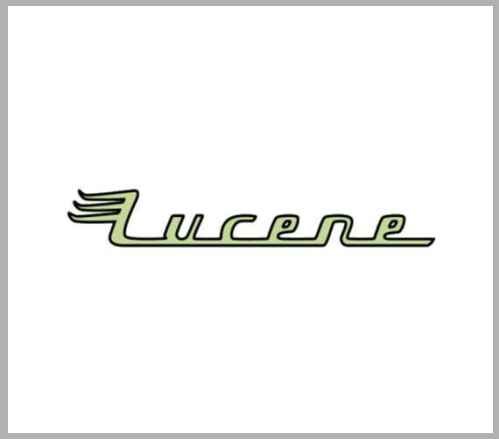
# OOZIE




- Oozie is a workflow / Coordination system that you can use to manage Apache Hadoop Jobs.
- Oozie Server – a web application that runs in a Java Servlet Container.
- This Server supports reading and executing Workflows, Coordinators and Bundles definitions.

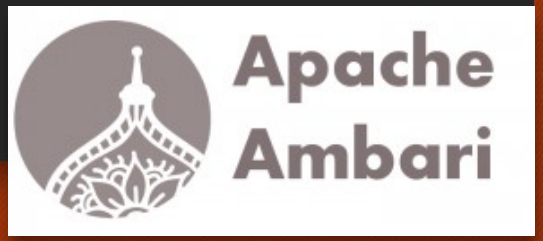
# APACHE SOLR & LUCENE




- Apache Solr and Apache Lucene are the two services which are used for searching and indexing in Hadoop Ecosystem.
- Apache Lucene and Apache Solr are developed in Java.
- If Apache Lucene is the engine, Apache Solr is the car built around it. Solr is a complete application built around Lucene.

# Apache Ambari




- Ambari is an Apache Software Foundation Project which aims at making Hadoop ecosystem more manageable.
- It includes software for **provisioning, managing and monitoring** Apache Hadoop clusters.

The Ambari provides:

**Hadoop cluster provisioning**:

It gives us step by step process for installing Hadoop services across several hosts.

It also handles configuration of Hadoop services over a cluster.

**Hadoop cluster management:**

It provides a central management service for starting, stopping and re-configuring Hadoop services across the cluster.

**Hadoop cluster monitoring:**

For monitoring health and status, Ambari provides us a dashboard.

The **Amber Alert framework** is an alerting service which notifies the user, whenever the attention is needed. For example, if a node goes down or low disk space on a node, etc.

- Forwarders Indexers Search Heads

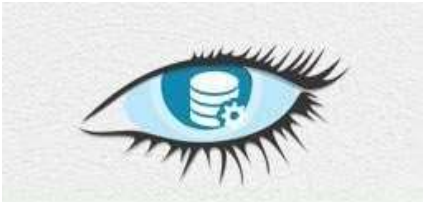
Splunk is a software platform to search, analyze and visualize the machine-generated data gathered from web-sites, applications, sensors, devices etc.

Splunk has three Major components

- **Developed by**: Splunk INC in the year 2014
- **Written in**: AJAX, C++, Python, XML
- **Current stable version**: Splunk 7.3

SPLUNK

# APACHE CASSANDRA



Apache Cassandra is an open source, and NoSQL database.

- It is scalable, fault-tolerant, and consistent.
- It is a column-oriented database.
- Its distribution design is based on Amazon's Dynamo and its data model on Google's Bigtable.
- Created at Facebook, it differs sharply from relational database management systems.
- Cassandra implements a Dynamo-style replication model with no single point of failure but adds a more powerful "column family" data model.
- Cassandra is being used by some of the biggest companies such as Facebook, Twitter, Cisco, Rackspace, ebay, Twitter, Netflix, and more.

# MongoDB

- MongoDB is an open-source, cross-platform, document-oriented database system which is written in C++ programming language.
- It works on the concept of Collection and document.
- MongoDB is Schemaless.
- MongoDB provides high performance, high availability, and easy scalability.






# TABLEAU




- Tableau is founded in 2003 by Chris, Christian and Pat.
- Tableau focuses on Visual Analysis.
- Tableau is a Business Intelligence tool.
- Tableau can be connected to any Database, Spreadsheets, Access Data Warehouses, Cloud-based Data Stores/Databases, Cloud Applications.
- Tableau Products
  - Tableau Desktop
  - Tableau Online
  - Tableau Server
  - Tableau Reader
  - Tableau Public



plotly

# PLOTLY

- Mainly **used for** creating Graphs faster and more efficient.
- API libraries for **Python, R**, **MATLAB, Node.js, Julia,** and **Arduino** and a **REST API.**
- **Plotly** can also be used to style **Interactive Graphs** with **Jupyter notebook.**
  - **Developed by**: Plotly in the year 2012
  - **Written in**: JavaScript
  - **Current stable version**: Plotly 1.47.4
- Companies Using Plotly: Paladins, bitbank.

# PRESTO

- **Presto** is an open source **Distributed SQL Query Engine** for running **Interactive Analytic Queries** against data sources of all sizes ranging from Gigabytes to Petabytes. Presto allows querying data in **Hive**, **Cassandra**, **Relational Databases** and **Proprietary Data Stores.**
- **Developed by**: Apache Foundation in the year 2013.
- **Written in**: JAVA
- **Current stable version**: Presto 0.22
- Companies Using **Presto**: Repro, Netflix, Airbnb, Checkr, facebook




- **RapidMiner** is a Centralized solution that features a very powerful and robust Graphical User Interface that enables users to Create, Deliver, and maintain Predictive Analytics. It allows creating very Advanced Workflows, Scripting support in several languages.

- **Developed by**: RapidMiner in the year 2001
- **Written in**: JAVA
- **Current stable version**: RapidMiner 9.2
- Companies Using **RapidMiner**: BCG Boston Consulting Group, Slalom, Domino's Vivint.SmartHome, InFocus.

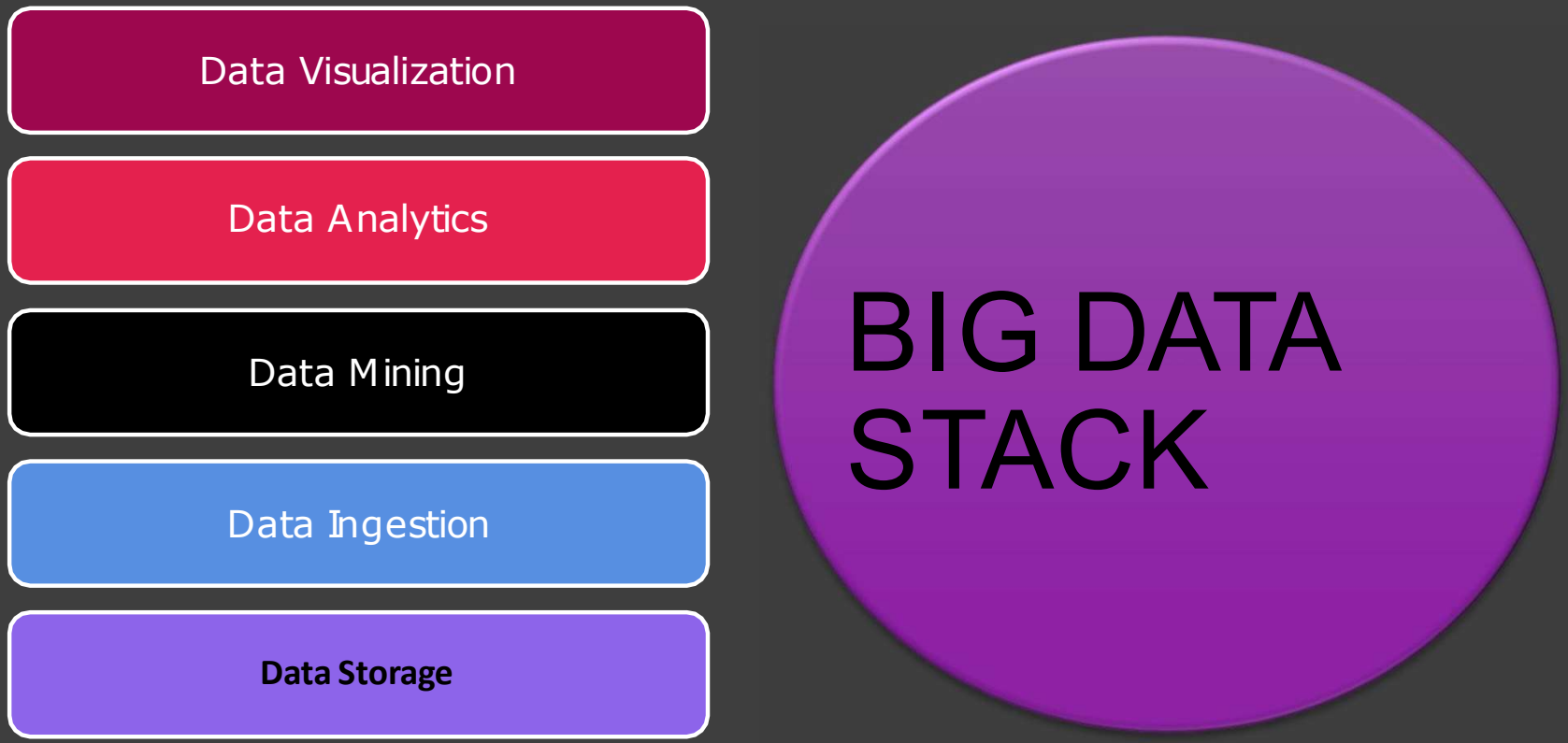


elasticsearch

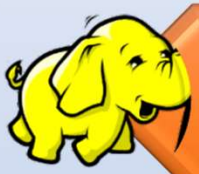
- **Elasticsearch** is a Search Engine based on the Lucene Library. It provides a Distributed, MultiTenant-capable, Full-Text Search Engine with an HTTP Web Interface and Schema-free JSON documents.

- **Developed by**: Elastic NV in the year 2012.
- **Written in**: JAVA
- **Current stable version**: ElasticSearch 7.1

- Companies Using **Elasticsearch**: Accenture, Stack Overflow, Netflix, LinkedIn


Data Visualization
Data Analytics
Data Mining
Data Ingestion
Data Storage
BIG DATA STACK

# Data Storage



Hadoop



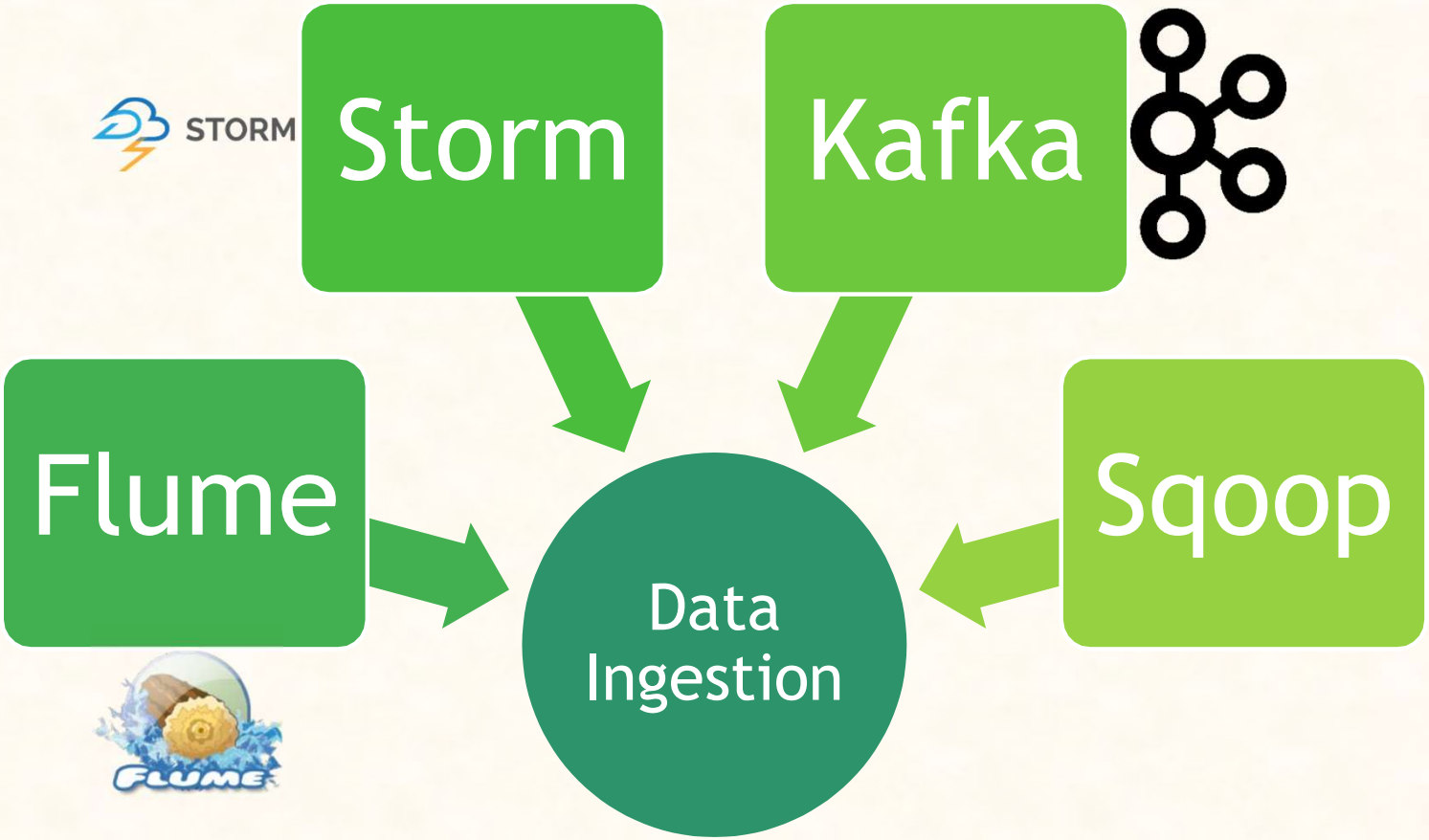

HBASE




Cassandra




MongoDB


STORM
Storm
Kafka
Flume
Sqoop
Data Ingestion
FLUME


rapidminer
Rapid Miner
Data Mining
presto
Presto
ElasticSearch
elasticsearch


Pig
Apache Drill
APACHE DRILL
Spark
Data Analytics
splunk>
Splunk
HIVE
Hive

# DATA VISUALIZATION

Tableau

Plotly

# Big Data Distribution Packages

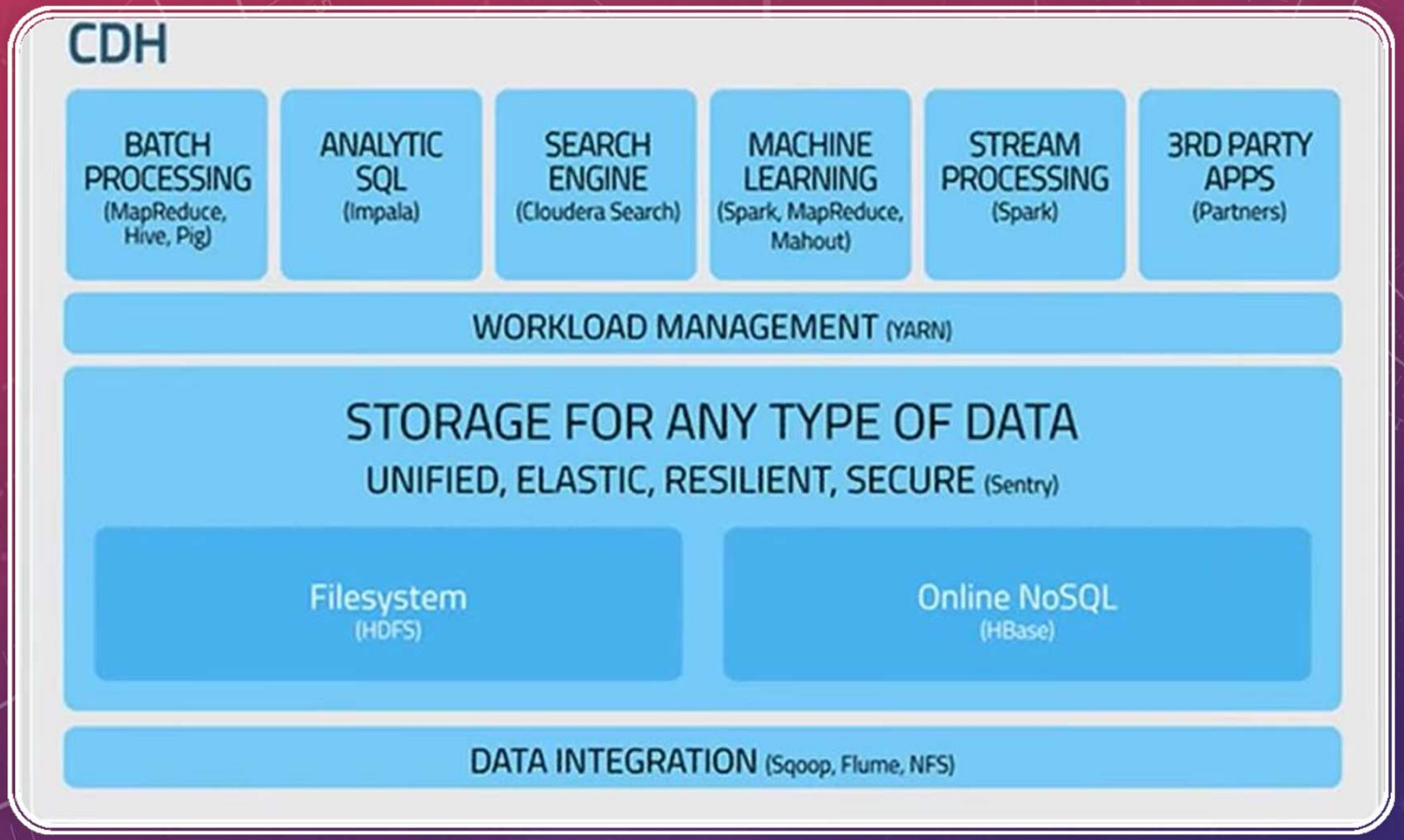
Cloudera

Hortonworks

MapR

# Cloudera Distributions

- First vendor to offer Hadoop as a package leader in the industry.
- First to offer SQL for Hadoop with its IMPALA query engine.
- CDH distribution contains Apache the open source components.
- Additions include user interface, Security and interfaces for integration with third party applications.
- Cloudera provides a proprietary Cloudera Manager for easy, fast, secure deployment and Management/Monitoring Services.



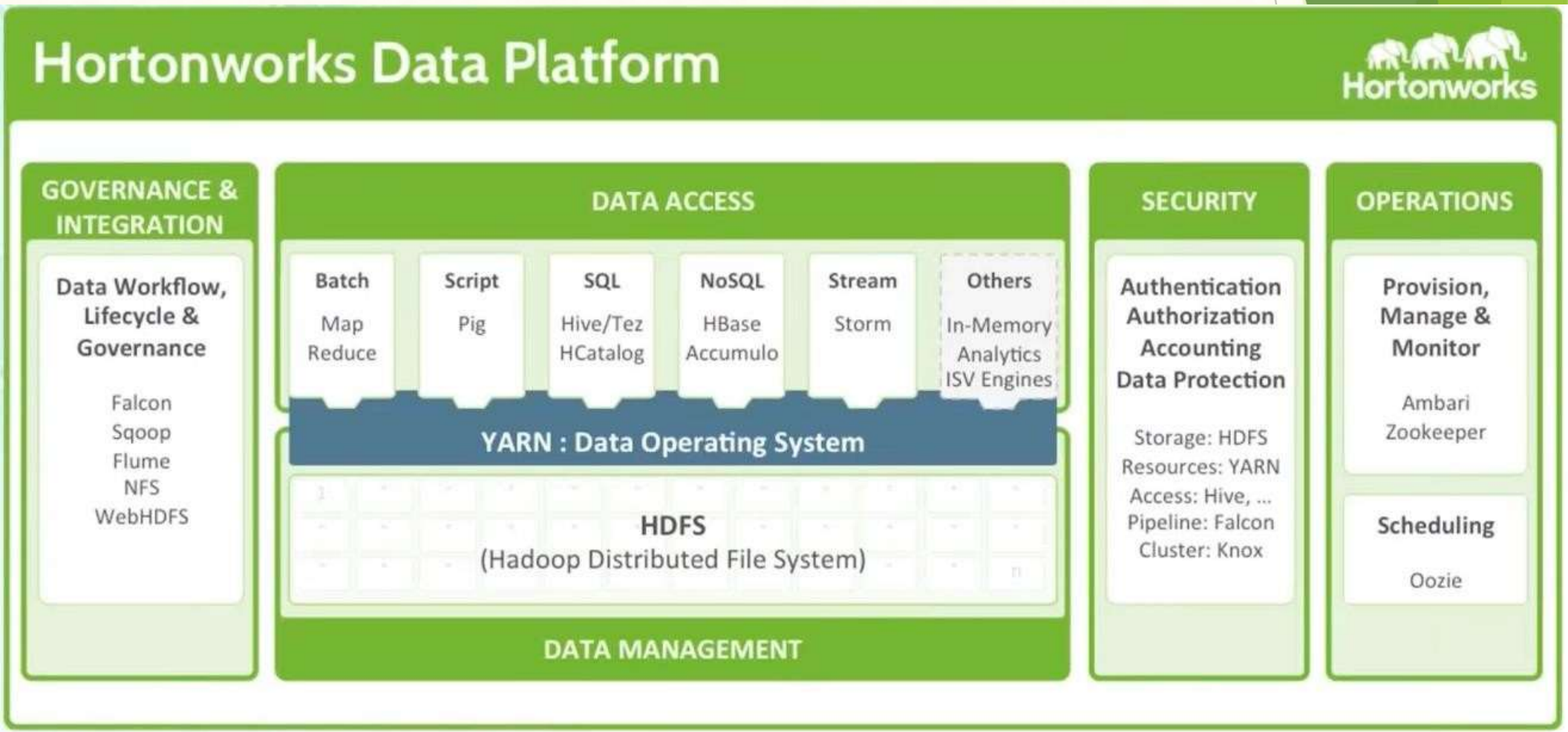

# CLOUDERA DISTRIBUTIONS

# CLOUDERA DISTRIBUTIONS PRODUCTS

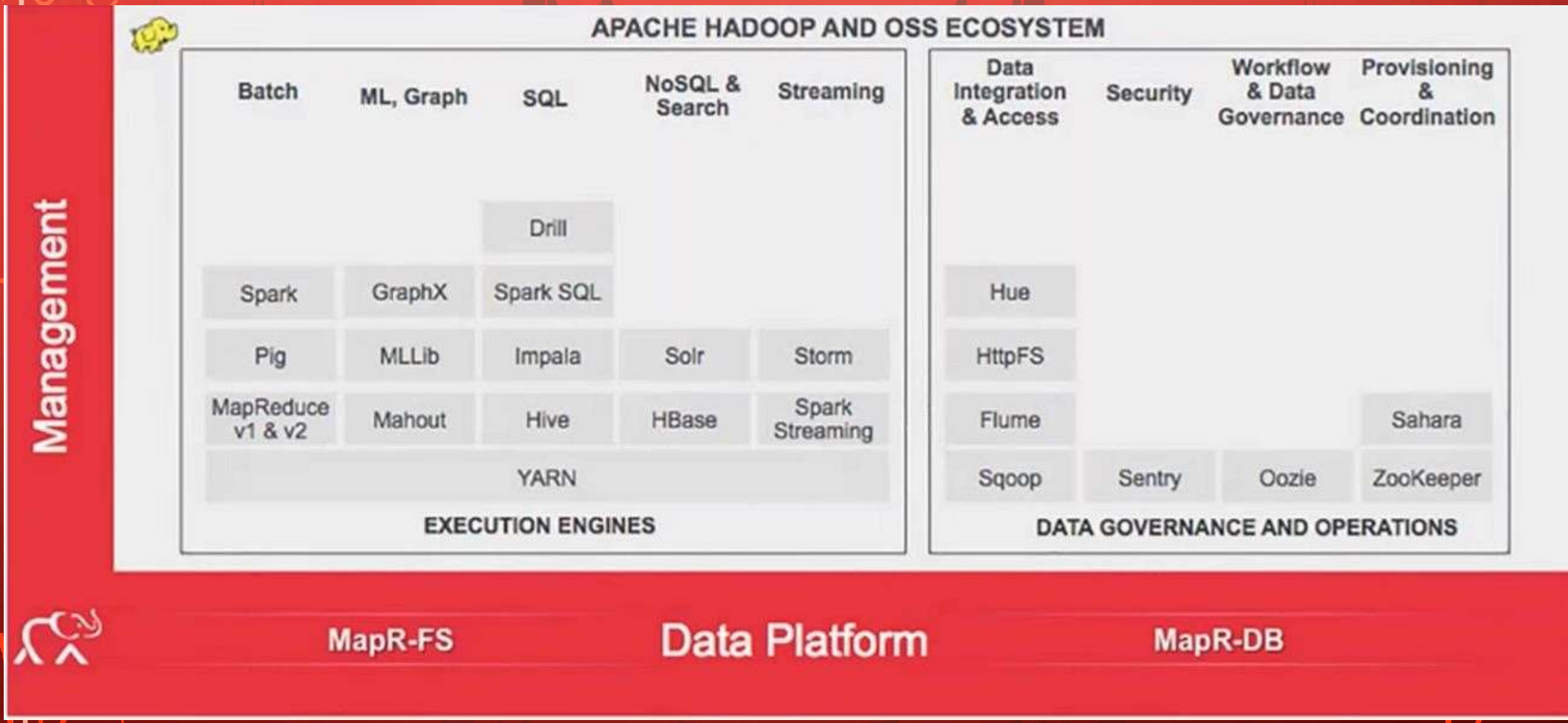
1. **Cloudera Express :** Cloudera Express is a free download that combines CDH with Cloudera Manager, which provides robust cluster management capabilities like deployment, centralized administration, monitoring and diagnostic tools.
2. **Cloudera Enterprise:** Cloudera Enterprise includes CDH with advanced system management and data management tools plus dedicated support from Cloudera
3. **Cloudera Director:** Cloudera Director extends Cloudera's enterprise data hub architecture to the cloud, without compromising on security, management and governance.

# Hortonworks

- Founded in 2011, Hortonworks has quickly emerged as one of the leading vendors of Hadoop.



# MAPR


APACHE HADOOP AND OSS ECOSYSTEM
Batch
ML, Graph
SQL
NoSQL & Search
Streaming
Drill
Spark
GraphX
Spark SQL
Pig
MLLib
Impala
Solr
Storm
MapReduce v1 & v2
Mahout
Hive
HBase
Spark Streaming
YARN
EXECUTION ENGINES
Data Integration & Access
Security
Workflow & Data Governance
Provisioning & Coordination
Hue
HttpFS
Flume
Sahara
Sqoop
Sentry
Oozie
ZooKeeper
DATA GOVERNANCE AND OPERATIONS
Management
MapR-FS
Data Platform
MapR-DB

# Cloudera Vs MapR Vs HortonWorks

| Features | Cloudera | MapR | HortonWorks |
|---|---|---|---|
| **License** | Commercial | Commercial | Open-Source |
| **Community** | Yes | Yes | Yes |
| **SQL** | Impala | Hive, Drill, Shark | Stinger |
| **MapReduce** | Yes | Yes | Yes |
| **Unique Features** | Cloudera Manager | Direct Access NFS | 100% open-source |
| **App Development** | Not Built-in | Not Built-in | Not Built-in |
| **Operating System** | LINUX | LINUX | LINUX |
| **Free to use** | 60 day Trial | M3 free addition | 100% free |
| **Revenue** | 56% of total big data | 23% of total big data | 18% of total big data |